

Polynomial visibility has density one

An elementary bounded-gcd argument

September 4, 2026

Abstract

For every nonzero integral polynomial with at least two distinct complex roots, almost every positive lattice point is visible along its rational vertical dilates. The proof has two ingredients: the gcd of a polynomial value and an independent height is bounded outside a set of arbitrarily small upper density; and every fixed proper ratio of polynomial values occurs at a set of arguments of density zero. Both ingredients admit elementary proofs. No uniform estimate over varying ratios is needed.

1 Statement and reduction

For $F \in \mathbb{Z}[x]$, call $(a, h) \in \mathbb{Z}_{>0}^2$ *visible* if there is $t \in \mathbb{Q}$ with $h = tF(a)$ and a is the least positive integer u for which $tF(u)$ is a positive integer. Put

$$V_F(N) = \#\{(a, h) \in [1, N]^2 \cap \mathbb{Z}^2 : (a, h) \text{ is visible}\}.$$

Theorem 1. *Let $F \in \mathbb{Z}[x] \setminus \{0\}$ have at least two distinct roots in $\mathbb{C}$. Then*

$$\lim_{N \rightarrow \infty} \frac{V_F(N)}{N^2} = 1.$$

This is the nonzero version of the Visibility Density Conjecture of Lobsenz–Phillips [2], extending the origin-passing formulation of Chaubey–Pandey [1]. Lobsenz–Phillips prove the assertion for proper powers of polynomials with at least two distinct roots; their revised paper also records the resolution of the case $F(0) = 0$ by Chaubey, Pandey, and Regavim. The argument below treats all nonzero polynomials under the stated root hypothesis. The explicit exclusion of $F = 0$ is necessary: otherwise every complex number is a root, but no positive lattice point is visible.

Replacing F by $-F$ leaves visibility unchanged, by replacing t by $-t$. We may therefore assume that its leading coefficient is positive. There is A_F such that, whenever $a > A_F$,

$$F(a) > \max(0, F(1), \dots, F(a-1)). \quad (1)$$

Indeed, a nonconstant polynomial with positive leading coefficient is eventually strictly increasing and tends to infinity. The columns $a \leq A_F$ contain only $O_F(N)$ points.

For $a > A_F$ set $g = \gcd(F(a), h)$. If (a, h) is invisible, its unique possible scale is $t = h/F(a)$, and some $1 \leq u < a$ satisfies $hF(u)/F(a) \in \mathbb{Z}_{>0}$. Consequently

$$F(a) \mid hF(u), \quad 0 < F(u) < F(a).$$

Since $F(a)/g$ and h/g are coprime, $F(a)/g$ divides $F(u)$. Thus there is an integer r such that

$$F(u) = \frac{r}{g} F(a), \quad 1 \leq r < g. \quad (2)$$

We will show that g can first be bounded at arbitrarily small density cost, and then handle the resulting finite list of ratios.

2 The gcd cutoff

Lemma 2. *For every nonzero $F \in \mathbb{Z}[x]$ and every $\varepsilon > 0$ there is an integer $M \geq 1$ such that, for every $N \geq 1$,*

$$\frac{1}{N^2} \#\{1 \leq a, h \leq N : \gcd(|F(a)|, h) > M\} < \varepsilon.$$

Proof. Write $d = \deg F$. Choose an integer $K \geq 2$ so large that every prime for which F is the zero polynomial modulo that prime is at most K . This excludes only finitely many primes, since F has a nonzero coefficient. For any prime $p > K$, the congruence $F(a) \equiv 0 \pmod{p}$ has at most d residue classes. Hence

$$\#\{1 \leq a, h \leq N : p \mid F(a), p \mid h\} \leq d \left(\frac{N}{p} + 1 \right) \frac{N}{p}.$$

Any prime dividing h is at most N , and for $p \leq N$ we have $N/p + 1 \leq 2N/p$. Summing over $K < p \leq N$ and enlarging the sum to all integers gives the bound, uniform in N ,

$$\frac{\#\{(a, h) \in [1, N]^2 : \exists p > K, p \mid \gcd(|F(a)|, h)\}}{N^2} \leq 2d \sum_{n > K} \frac{1}{n^2} \leq \frac{2d}{K}. \quad (3)$$

For any fixed positive integer E , the proportion of heights divisible by p^E is at most p^{-E} . Outside a set of pairs of proportion at most $\sum_{p \leq K} p^{-E}$, none of these small-prime powers divides h . If neither exceptional event occurs, then

$$\gcd(|F(a)|, h) \leq M := \prod_{p \leq K} p^{E-1}.$$

Choose K with $2d/K < \varepsilon/2$, and then E with $\sum_{p \leq K} p^{-E} < \varepsilon/2$. Combining the two exceptional sets proves the claim. This also covers $F(a) = 0$, since $\gcd(0, h) = h > 0$ and the same prime-divisor argument applies. $\square$

3 Fixed ratios are sparse

Lemma 3. *Let $F \in \mathbb{R}[x]$ have degree $d \geq 2$, positive leading coefficient, and at least two distinct complex roots. For each $q \in (0, 1)$, the set*

$$S_q = \{a \in \mathbb{Z}_{>0} : \exists u \in \mathbb{Z}_{>0}, u < a, F(u) = qF(a)\}$$

has natural density zero.

Proof. Write

$$F(x) = cx^d + bx^{d-1} + O_F(x^{d-2}), \quad \alpha = q^{1/d} \in (0, 1), \quad \beta = \frac{b(\alpha - 1)}{dc}.$$

For any fixed $\eta > 0$, the polynomials $F(\alpha x + \beta \pm \eta) - qF(x)$ have degree $d - 1$ and leading coefficients $\pm dc\alpha^{d-1}\eta$. Consequently, for large a ,

$$F(\alpha a + \beta - \eta) < qF(a) < F(\alpha a + \beta + \eta).$$

Along solutions with $a \rightarrow \infty$ we have $u \rightarrow \infty$. Eventual strict monotonicity of F therefore places u between these two arguments. As η is arbitrary, this proves

$$u = \alpha a + \beta + o_{F,q}(1). \quad (4)$$

Suppose first that α is irrational. Fix $L \geq 1$ and set

$$\delta_L = \min_{1 \leq j \leq L} \text{dist}(j\alpha, \mathbb{Z}) > 0.$$

Beyond a fixed threshold, the error in (4) has absolute value less than $\delta_L/3$. Two such solutions with $1 \leq a' - a \leq L$ would give

$$\text{dist}((a' - a)\alpha, \mathbb{Z}) \leq |(u' - u) - \alpha(a' - a)| < 2\delta_L/3,$$

a contradiction. Thus every block of $L + 1$ consecutive sufficiently large integers contains at most one member of S_q . Its upper density is at most $1/(L + 1)$; letting $L \rightarrow \infty$ proves the assertion.

Suppose next that α is rational. Choose $D \geq 1$ such that $D\alpha \in \mathbb{Z}$. Then $u - \alpha a \in D^{-1}\mathbb{Z}$. Equation (4) shows that, if arbitrarily large solutions exist, necessarily $\beta \in D^{-1}\mathbb{Z}$ and all sufficiently large solutions satisfy $u = \alpha a + \beta$ exactly. If infinitely many solutions existed, the polynomial

$$F(\alpha x + \beta) - qF(x)$$

would have infinitely many zeros, and so would vanish identically. Let R be the finite nonempty set of complex roots of F , and let $T(z) = \alpha z + \beta$. This identity implies $T(R) \subseteq R$. As T is injective and R finite, $T(R) = R$. But the diameter satisfies

$$\text{diam}(R) = \text{diam}(T(R)) = \alpha \text{diam}(R).$$

Since $0 < \alpha < 1$, the diameter is zero, contrary to the existence of two distinct roots. In this case S_q is in fact finite. $\square$

4 Completion of the proof

Proof of Theorem 1. Fix $\varepsilon > 0$ and choose M from Lemma 2. Let

$$S_M = \bigcup_{2 \leq g \leq M} \bigcup_{1 \leq r < g} S_{r/g}.$$

This is a finite union of sets of density zero by Lemma 3, so $\#(S_M \cap [1, N]) = o_{F,M}(N)$. By (2), every invisible point in $[1, N]^2$ either has $a \leq A_F$, has $\gcd(F(a), h) > M$, or has $a \in S_M$. It follows that

$$\limsup_{N \rightarrow \infty} \frac{N^2 - V_F(N)}{N^2} \leq 0 + \varepsilon + \lim_{N \rightarrow \infty} \frac{N \#(S_M \cap [1, N])}{N^2} = \varepsilon.$$

Since ε is arbitrary and the left side is nonnegative, $N^2 - V_F(N) = o(N^2)$. $\square$

Remark 4. *The order of limits is essential: first fix an error tolerance and its gcd cutoff M , then let $N \rightarrow \infty$. The proof requires no uniform control of $S_{r/g}$ as g grows. In particular it does not assert an unproved average-gcd bound for two polynomial values.*

Formal verification. The accompanying Lean 4.28.0 project proves the integer-polynomial theorem as `PolynomialVisibility.visibility_density_one`, using `mathlib v4.28.0`. The complete proof has no admitted steps or additional axioms; its dependencies are only the standard logical foundations `propext`, `Classical.choice`, and `Quot.sound`.

References

- [1] S. Chaubey and A. K. Pandey, *On the density of visible lattice points along polynomials*, 2021, [arXiv:2109.08431](#).
- [2] A. Lobsenz and T. Phillips, *Lattice point visibility along powers of polynomials*, 2026, [arXiv:2604.23050v3](#).